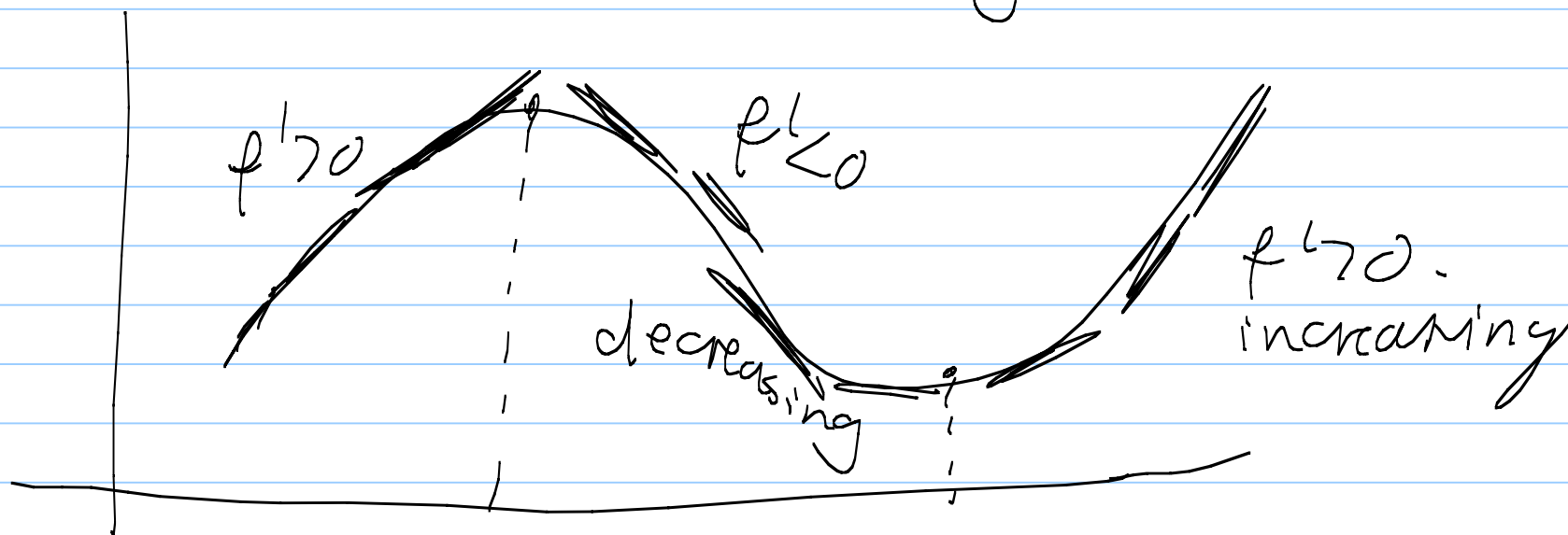


Unit 4.3 How derivatives affect the shape of a graph

Note Title

2012/04/20

What does f' say about f ?



Def

- 1) A function f is increasing on an interval I if
$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2) \quad \forall x_1 \text{ and } x_2 \text{ in } I$$
- 2) A function f is decreasing on an interval I if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$ for all x_1 and x_2 in I .

Increasing / Decreasing test

a) If $f'(x) > 0$ on an interval I then f is increasing on that interval

b) If $f'(x) < 0$ on an interval I , Then f decreasing on that interval

Example Find The intervals on which

$f(x) = xe^x$ is decreasing.

Sol: $f'(x) = e^x + xe^x$

$$\Rightarrow f'(x) < 0$$

$$\Rightarrow e^x(1+x) < 0$$

e^x will never be < 0



$$\Rightarrow 1+x < 0$$

$$\Rightarrow x < -1$$

The function is decreasing
on $(-\infty, -1)$.

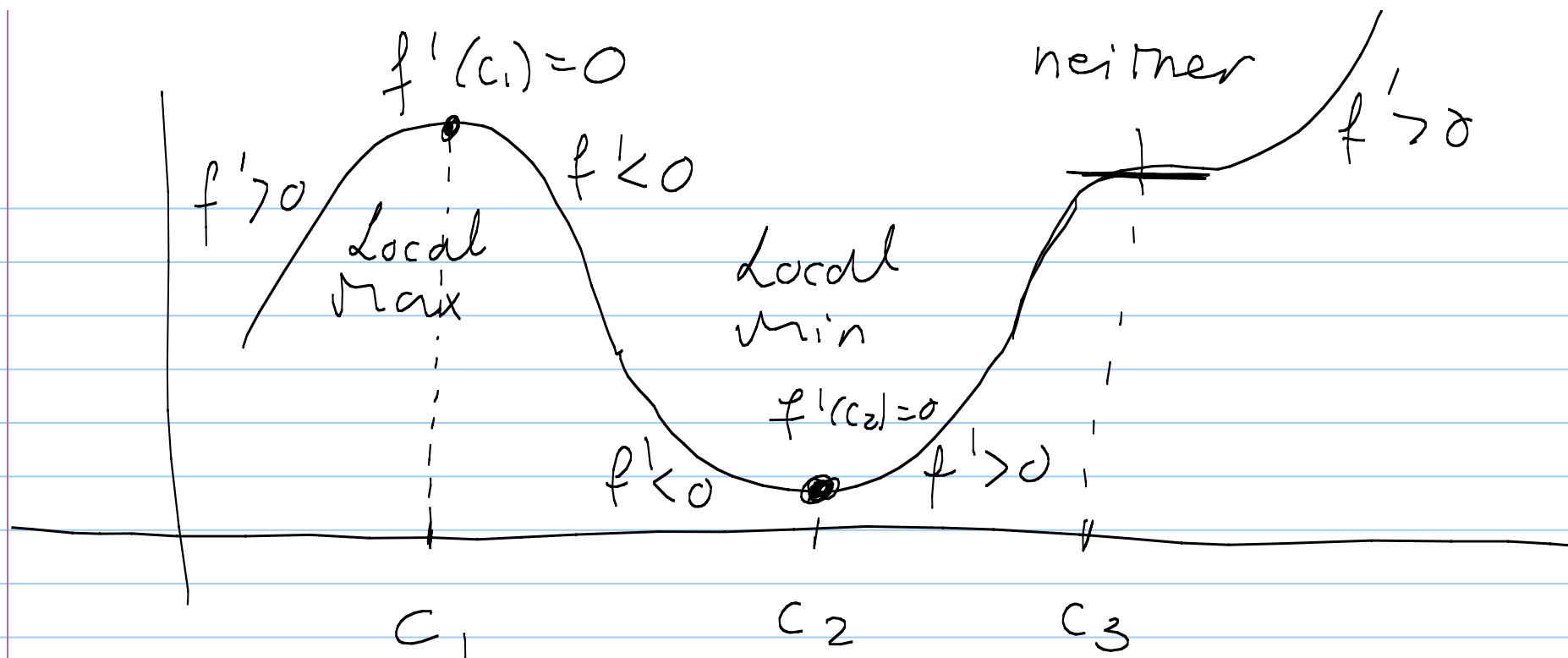
The First derivative test.

Suppose that c is a critical number of a continuous function f .

a) If f' changes from positive to negative at c ,
Then f has a local maximum at c .

b) If f' changes from negative to positive at c , then f has a local min at c .

c) If f' does not change sign at c , then f has no local max or min at c .



Example Find local
extremes of

$$f(x) = -x^5 + 15x^3$$

- As f is a polynomial f is continuous on $\mathbb{R}$.

- $$\begin{aligned} f'(x) &= -5x^4 + 45x^2 \\ &= -5x^2(x^2 - 9) \\ &= -5x^2(x-3)(x+3) \end{aligned}$$

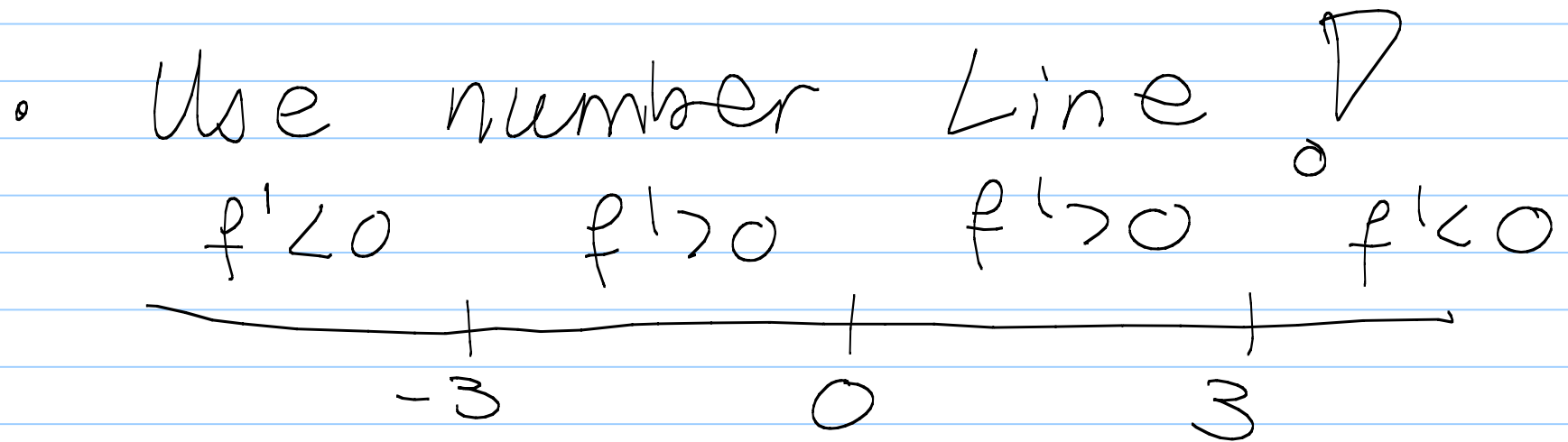
$\therefore f$ is differentiable on $\mathbb{R}$.

- $f'(x) = 0$

$$\Rightarrow x=0 \text{ or } x=3 \text{ or } x=-3$$

The critical numbers are

$$x=0, \quad x=3, \quad x=-3$$



• f is decreasing on $(-\infty, -3)$
and increasing on $(-3, 3)$
which means that
 $f(-3) = -162$ is a local Min.

• f is increasing on $(0, 3)$ and
decreasing on $(3, \infty)$ which
means $f(3) = 162$ is a local
Max.

Example

- ① Give intervals on which the function $f(x) = x^2(x-1)^2$ is increasing. And find local extremes.

Sol:

- differentiate

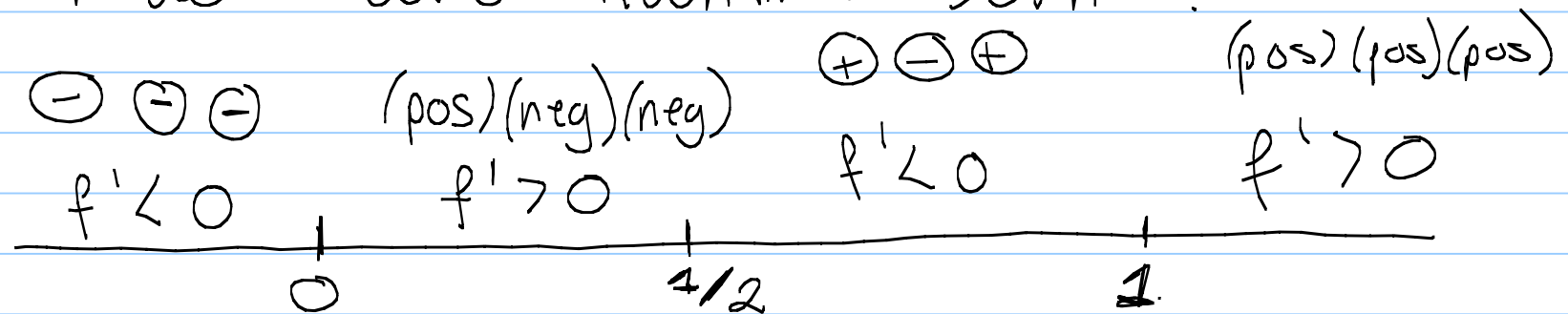
$$\begin{aligned} f'(x) &= 2x(x-1)^2 + x^2 \cdot 2(x-1) \quad \leftarrow \text{product rule.} \\ &= 2x(x-1) \left((x-1) + x \right) \\ &= 2x(x-1) (2x-1) \end{aligned}$$

- critical numbers $f'(x) = 0$

$$2x(x-1)(2x-1) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 1 \text{ or } x = \frac{1}{2}$$

- Must use number line.



- $f(x)$ is increasing when $f'(x) > 0$.

$\Rightarrow f(x)$ is increasing on interval $(0, \frac{1}{2})$ and $(1, \infty)$.

• Local extremes:

• f' changes from neg to pos at 0 and 1

$\Rightarrow$ Local Min at

$$x=0, \quad f(0)=0$$

$$\text{and } x=1, \quad f(1)=0$$

• f' changes from pos to neg at $\frac{1}{2}$

$\Rightarrow$ Local Max at $x=\frac{1}{2}$, function value
 $f(\frac{1}{2}) = \frac{1}{4}(-\frac{1}{2})^2 = \frac{1}{16}$

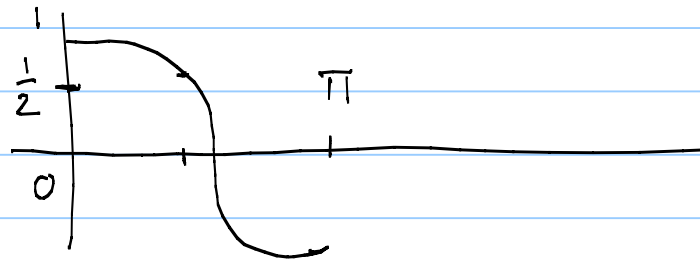
② Consider $f(x) = x - 2\sin x$ $x \in (0, \pi)$.
Find The intervals on which the
function $f(x)$ is decreasing and
find local extremes.

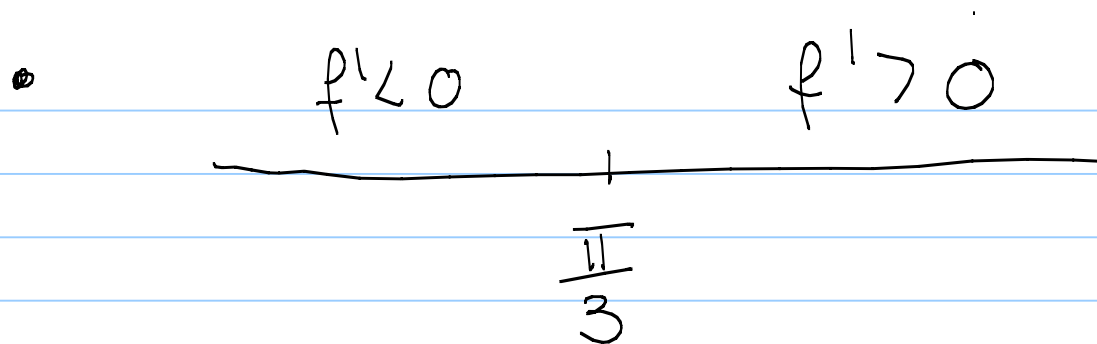
Sol:

- $f'(x) = 1 - 2\cos x$

- $f'(x) = 0 \Rightarrow \cos x = \frac{1}{2}$

$$\Rightarrow x = \frac{\pi}{3}$$

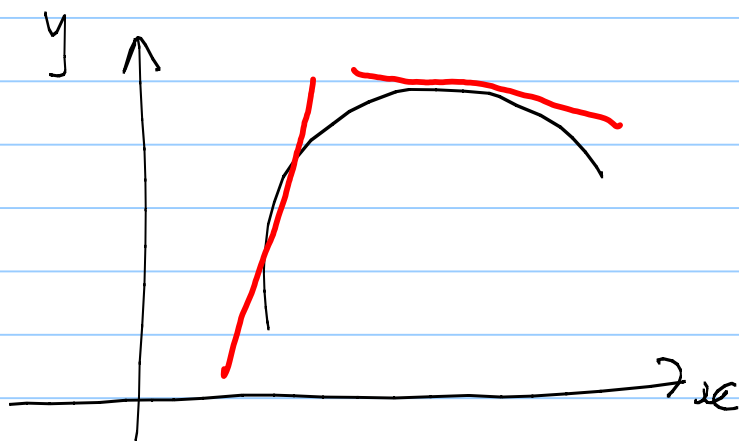




• $f(x)$ is decreasing when $f' < 0$
which is on the interval $(0, \frac{\pi}{3})$

• $f'(x)$ changes from neg to pos at
 $x = \frac{\pi}{3} \Rightarrow f(\frac{\pi}{3}) = 0$ is a
Local min.

p 292 What does f'' say about f ?



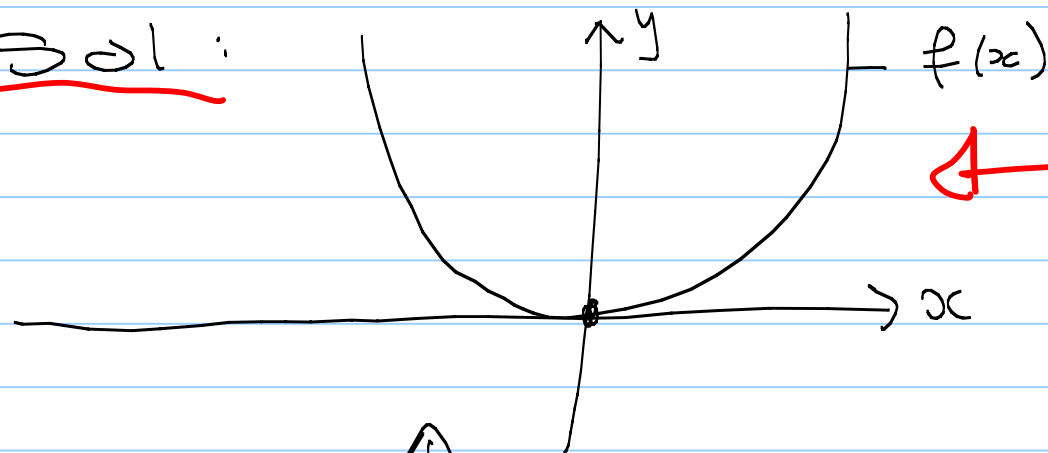
Definition p 293.

If the graph of f lies above all of its tangents on an interval I , then it is called concave upward on I .

If the graph of f lies below all of its tangents on I , it is called concave downward on I .

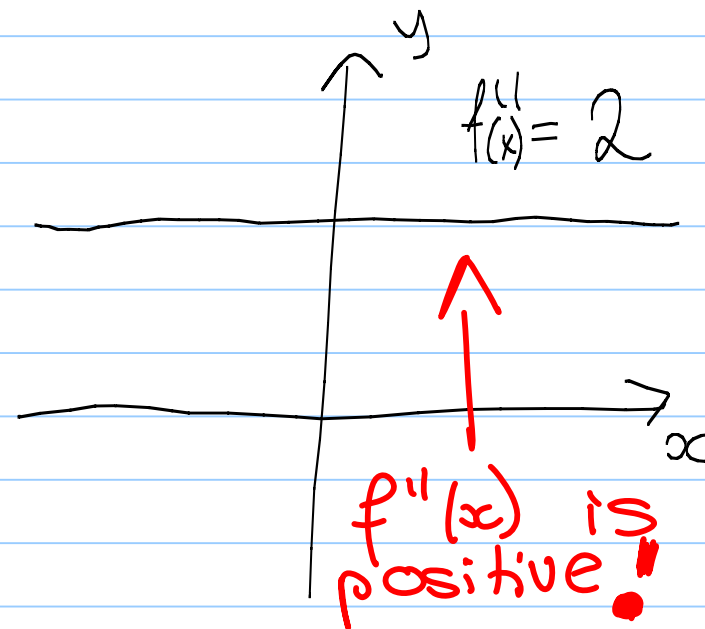
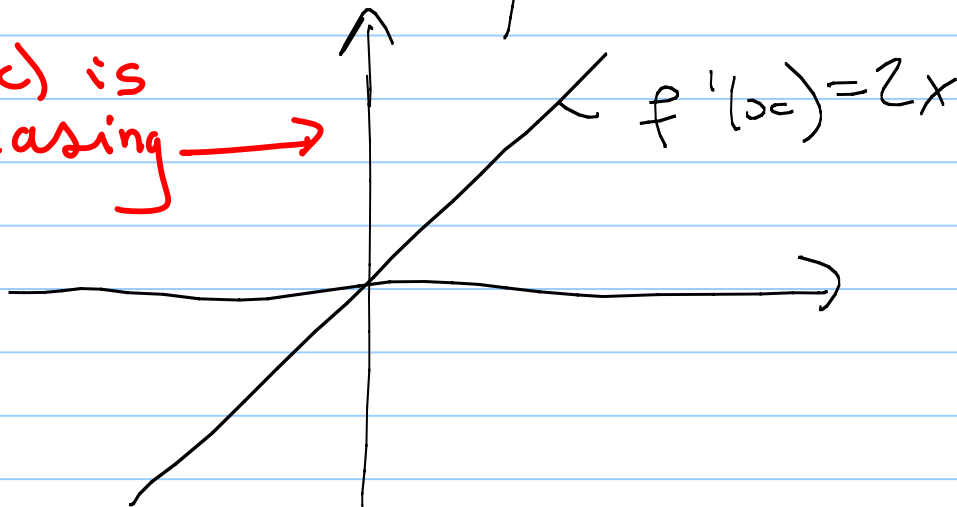
Example: Sketch $f(x) = x^2$, $f'(x)$ & $f''(x)$

Sol:

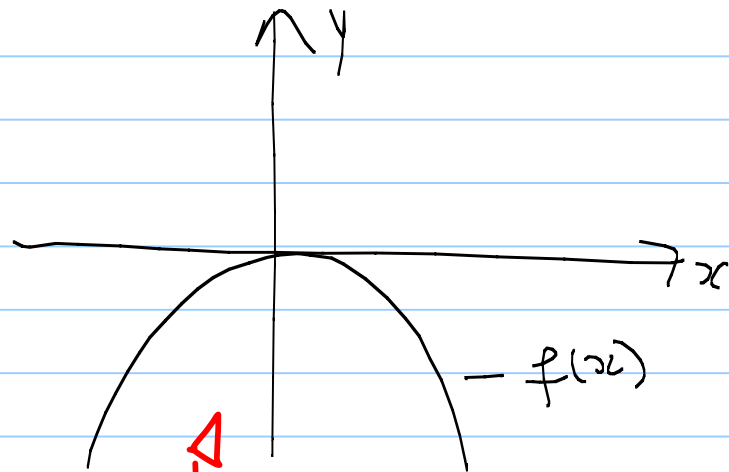


← concave up

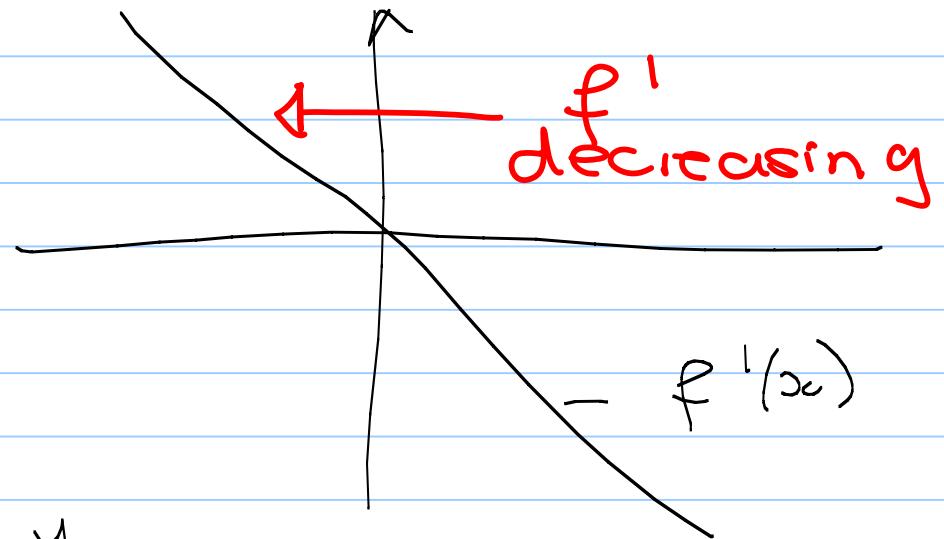
$f'(x)$ is increasing →



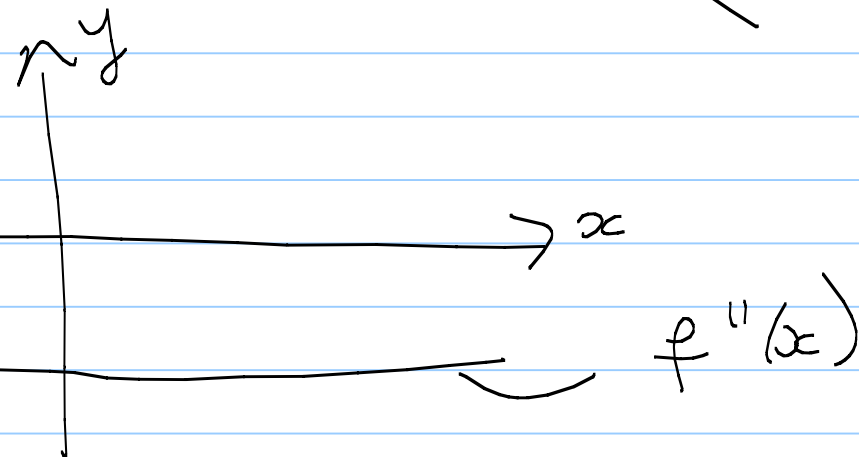
Example. Sketch $f(x) = -x^2$, $f'(x)$ & $f''(x)$



concave
down



$f'' < 0$
second derivative
neg!



Concavity test

MB.

a) If $f''(x) > 0$ for all x in I ,
Then the graph of f is
concave upward on I .

b) If $f''(x) < 0$ $\forall x$ in I ,
Then f is concave downward
on I .

Example: Find intervals where $f(x) = x^3$ is concave downward.

Sol:

$$f'(x) = 3x^2$$

$$f''(x) = 6x$$

$$f''(x) < 0 \Rightarrow 6x < 0 \Rightarrow x < 0$$

$f(x)$ is concave downward on $(-\infty, 0)$.



Definition. A point P on a curve $y = f(x)$ is called an inflection point if f is continuous there and the curve changes from concave upward to concave downward or concave downward to concave upward at P .

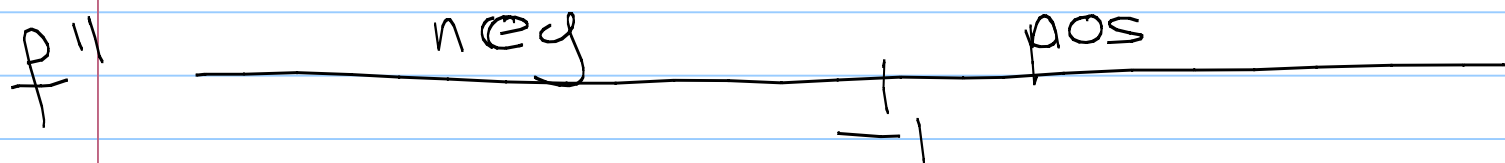


Example Find the inflection points of $f(x) = 1 + 3x + 3x^2 + x^3$

Sol: $f'(x) = 3 + 6x + 3x^2$

$$f''(x) = 6 + 6x = 6(1+x)$$

$$f''(x) = 0 \Rightarrow x = -1$$



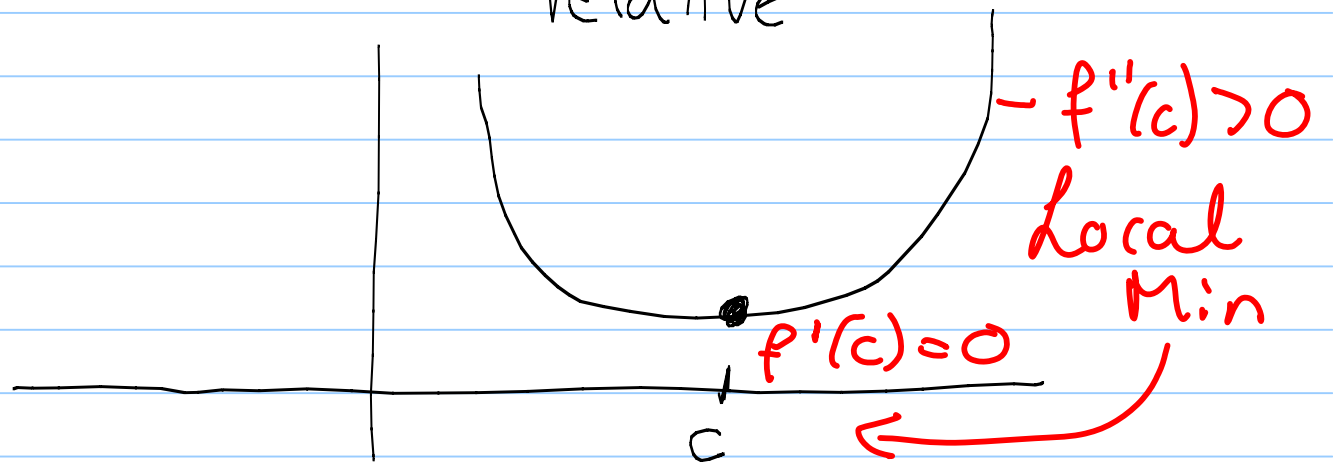
f is concave upward on $(-1, \infty)$ and concave downward on $(-\infty, -1)$.

$\therefore f$ has an inflection point at $x = -1$

The Second derivative test

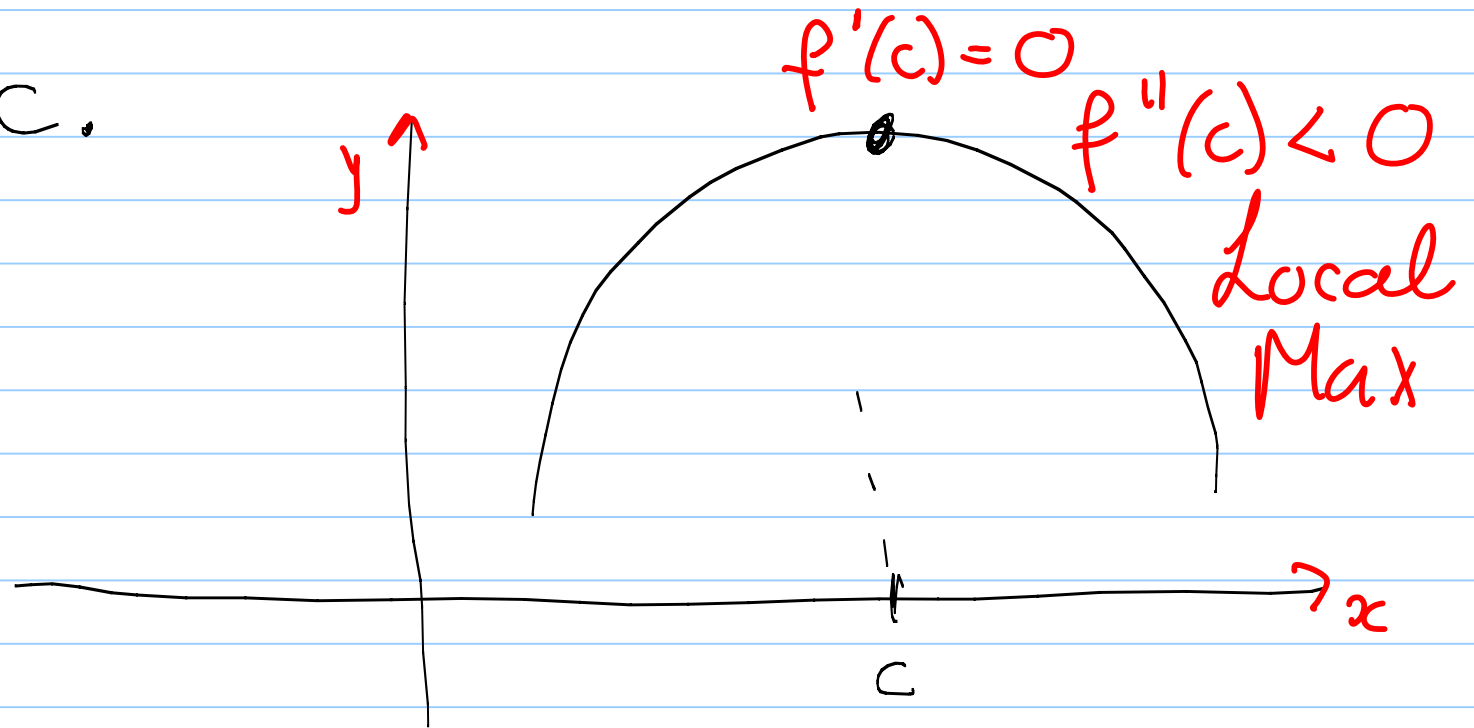
Suppose f'' is continuous near c

a) If $f'(c) = 0$ AND $f''(c) > 0$,
then f has a local/relative minimum
at c .



b) If $f'(c) = 0$ AND $f''(c) < 0$,

Then f has a local maximum
at c .



NOTE :

$$\text{If } f'(c) = 0 \text{ AND } f''(c) = 0$$

You cannot use the
second derivative test!

Use first derivative test.

Example Given

$$f(x) = x^2 e^{-x^2}$$

$$f'(x) = 2x e^{-x^2} (1 - x^2)$$

$$f''(x) = 2 e^{-x^2} (2x^4 - 5x^2 + 1)$$

and the critical values

are $x = -1$, $x = 0$ and $x = 1$.

Use The second derivative test to classify the local extremes.

Sol:

● $x = -1$ is critical point

$$\Rightarrow f'(-1) = 0$$

$$\text{and } f''(-1) = 2e^{-(-1)^2} (2(-1)^4 - 5(-1)^2 + 1)$$

$$= 2e^{-1} (2 - 5 + 1)$$

$$= 2e^{-1} (-2) = -4e^{-1} < 0$$

remember e^{-1} is pos $\neq$

$\Rightarrow$ By 2nd derivative test

f has a local max
at $x = -1$.

- at $x = 0$

$$f'(0) = 0$$

$$\begin{aligned} f''(0) &= 2e^{-0^2} (2(0)^4 - 5(0)^2 + 1) \\ &= 2(1)(1) = 2 > 0 \end{aligned}$$

$\Rightarrow f$ has Local Min at $x=0$

• at $x=1$

$$f'(1) = 0$$

$$f''(1) = 2e^{-1}(-2) = -4e^{-1} < 0$$

$\Rightarrow f$ has Local max at 1

Examples.

Find local extremes of

1] $f(x) = \frac{\ln x}{x}$ and

2] $f(x) = x e^{-3x}$

Solution of 1: $f(x) = \frac{\ln x}{x}$

$$f'(x) = -\frac{1}{x^2} \ln x + \frac{1}{x^2}$$

$$= \frac{1}{x^2} [1 - \ln x]$$

critical numbers $f'(x) = 0$

$$\Rightarrow 1 - \ln x = 0 \Rightarrow \ln x = 1 \Rightarrow x = e$$

$$\begin{aligned} f''(x) &= -\frac{1}{x^3} (1 - \ln x) + \frac{1}{x^2} \cdot \left(-\frac{1}{x}\right) \\ &= -\frac{1}{x^3} [2 - \ln x] \end{aligned}$$

$$f''(e) = -\frac{1}{e^3} [2 - \ln e]$$

$$= -e^{-3} < 0$$

By 2nd derivative test
f has local max at $x=e$.

Solution of 2: $f(x) = x e^{-3x}$.

$$f'(x) = e^{-3x} - 3x e^{-3x}$$

$$= e^{-3x} [1 - 3x]$$

critical numbers $f'(x) = 0$

$$e^{-3x} > 0 \quad \text{and} \quad 1 - 3x = 0$$

$$\Rightarrow x = 1/3$$

$$f''(x) = -3e^{-3x} (1-3x) + e^{-3x} (-3)$$

$$= -3e^{-3x} [1-3x+1]$$

$$= -3e^{-3x} (2-3x)$$

$$f''\left(\frac{1}{3}\right) = -3e^{-3\left(\frac{1}{3}\right)} \left[2-3\left(\frac{1}{3}\right)\right]$$

$$= -3e^{-1} [1] = -3e^{-1} < 0$$

$\Rightarrow f$ has a Local Max at $x = \frac{1}{3}$.

